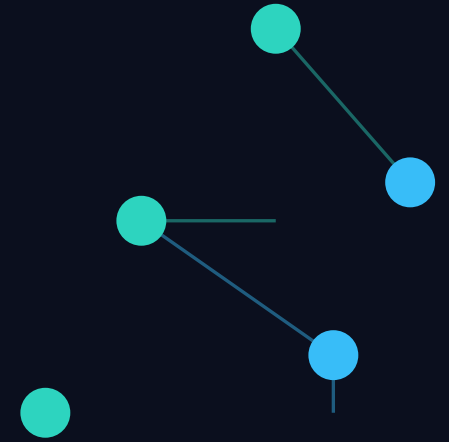




CHAINSILIENCE AI



Empowering SMEs and startups to foster resilience to global supply chain disruptions with AI.

Agentic supply-chain decision support combining LLM/NLP, RAG, knowledge graphs, a digital twin, simulation, optimization, and governance.

Team: University of Toronto · Ka Woon Ho, Ke Xin Cao, Gary Leng, Mo Xia, Andrew Li

Focus: SMEs · startups · semiconductor and electronics supply networks

Disruption signals arrive before operational context

● Taiwan earthquake

● US export restrictions

● Rare-earth shortages

● Shipping & port delays

● New tariffs

Manual workflow breaks at four points

High-volume reporting hides enterprise-relevant events.

- 1 Relevant events are detected too late
- 2 Unsupported reports create costly false alarms
- 3 Dependencies remain scattered across enterprise systems
- 4 News rarely produces a feasible operational response

A small timing gap can become a stoppage

60%

qualified wafer volume from one
supplier

12 days

usable inventory coverage

21 days

illustrative recovery range

Day 13

projected baseline shortage

Illustrative Orion Controls scenario: concentrated supply, short inventory coverage, and slow recovery create a time-critical enterprise decision.

From external signal to governed decision

Every transition stores confidence, provenance, timestamp, and version.

1

Detect & Verify

Normalize permitted sources and cross-check disruption claims

2

Map Exposure

Link events to enterprise parts, routes, products, and orders

3

Quantify Impact

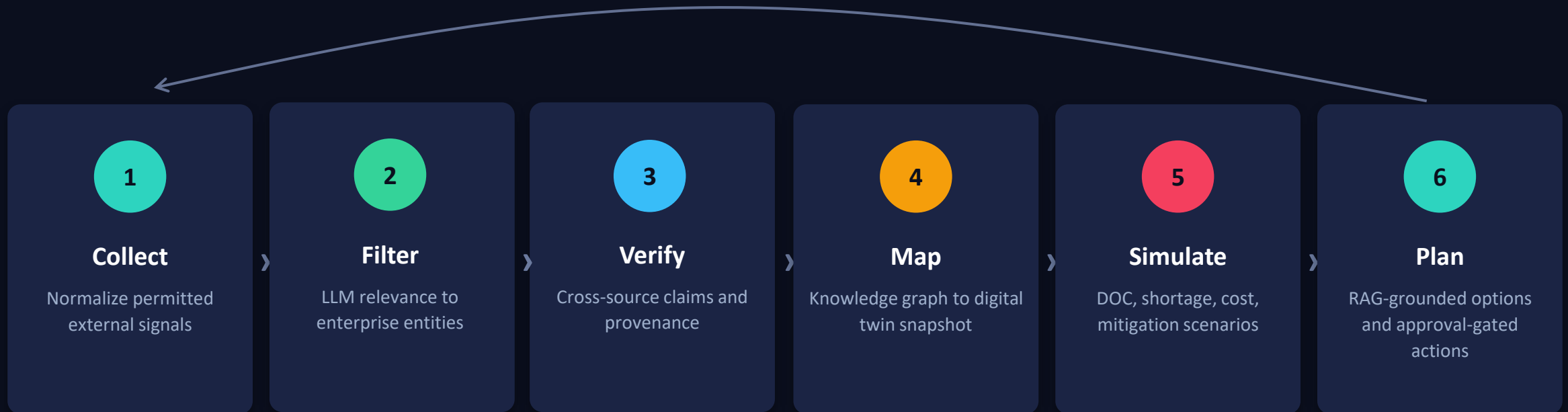
Deterministic metrics and Monte Carlo scenario analysis

4

Plan & Approve

RAG-grounded options, feasibility checks, and human authorization

One evidence-preserving intelligence loop



New confirmations, inventory changes, shipment data, and production results update the versioned twin while preserving before-and-after evidence.

Bounded agents turn news into governed action



Evidence enters the LLM before an answer

RAG retrieves authorized records; verification filters out unreliable, contradictory, or unsupported claims.

1

External evidence

Government advisories, feeds, trade notices, disclosures, trusted media, and cited passages

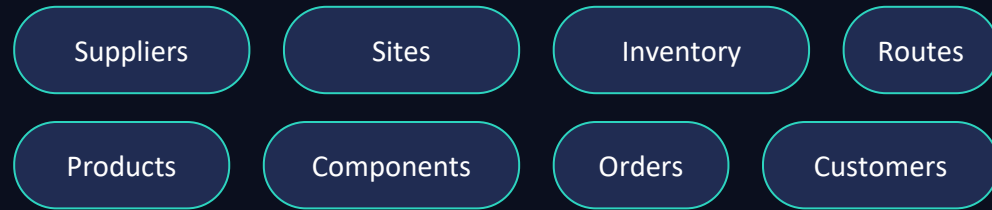
2

Enterprise context

Approved suppliers, substitutes, contracts, policies, capacities, routes, and prior responses

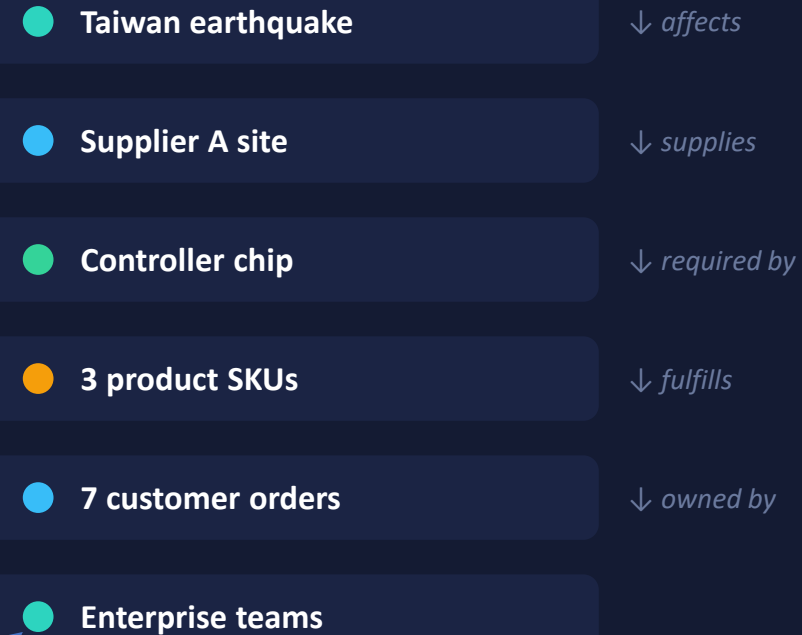
Knowledge graph turns relevance into exposure

Authorized enterprise data becomes a versioned graph plus time-indexed operational state.



State includes inventory, demand, capacity, lead time, routes, orders, priorities, source, and effective time.

Illustrative affected graph path



Edge

Node

Each component performs one accountable job

Language understanding, structured context, quantitative analysis, and authorization remain separate.

● LLM + NLP

Relevance, extraction, explanation

● RAG

Authorized evidence and constraints

● Digital twin

Versioned replica of enterprise entities

● Knowledge graph

Affected supply chain paths

● Simulation + optimization

Impact, feasibility, trade-offs

● Policy + human approval

Authorization and accountability

87/100

Critical Risk

Deterministic services calculate. The LLM interprets and explains score reasoning. Enterprise policy and authorized roles decide.

Operational impact becomes reproducible

Illustrative Orion Controls outputs from one frozen enterprise snapshot.

13

projected shortage day

12

days of inventory coverage

71%

Production stoppage probability

\$180K

mitigation cost estimate

3 weeks

illustrative recovery range

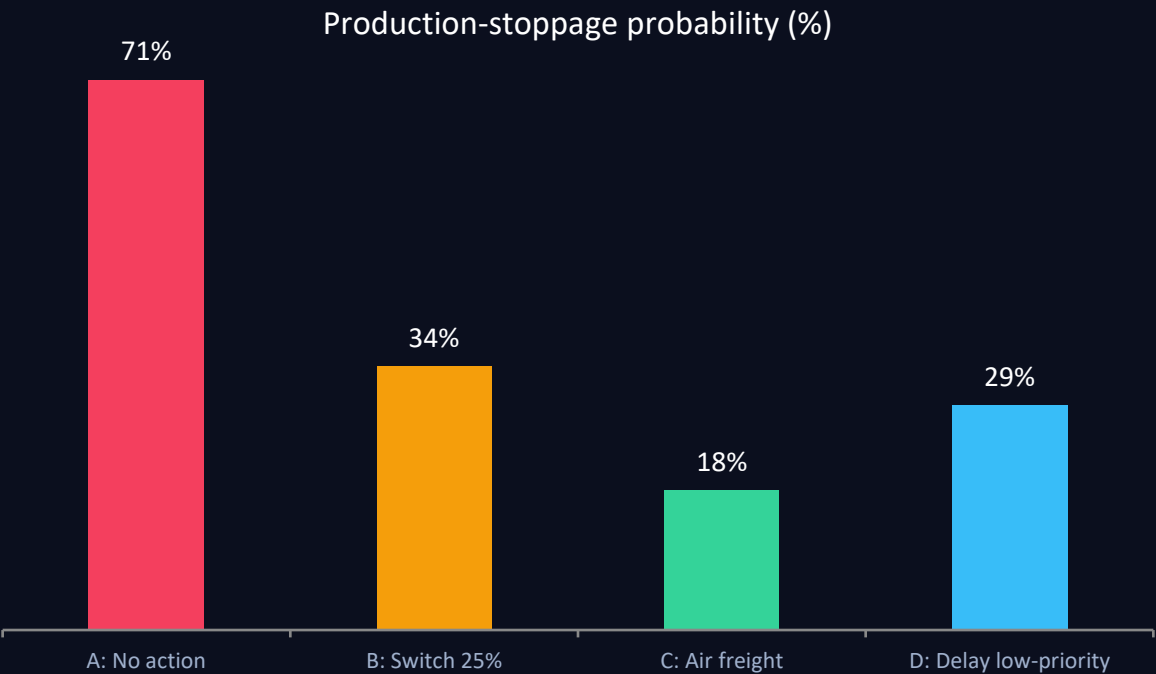
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customer orders on affected path

Method: deterministic inventory and cost calculations plus Monte Carlo simulation. The LLM explains outputs and reasoning behind the numbers; the LLM does not perform the calculations.

Candidate mitigation actions are grounded, checked, and compared

RAG provides tailored mitigation options and constraints; deterministic services validate options numerically.



Illustrative scenario simulation comparison

- B • Switch 25% to Supplier B**
\$180K estimate • 5-day changeover • qualification checked
- C • Use qualified air freight**
\$300K estimate • carrier • customs • capacity • cost checked
- D • Delay low-priority orders**
\$0 estimate • Priority • contract • service commitments checked

AI recommendations remain reviewable drafts

One reviewable recommendation package

Evidence

Verified sources · cited passages · affected graph path

AI-recommended mitigation strategies

Proposed actions · projected impact · cost · recovery time · financial impact

Human review and decision

Named approver · edits · approval status · deadline · audit history

AI-drafted supplier request email

“Please confirm facility status, available capacity, revised lead time, allocation policy, and next update time for Purchase Order 1042.”

Supplier/customer/executive drafts remain editable and approval-gated.

Policy gateway: Consequential actions require named authorization. MCP may expose typed tools, but authentication, validation, and approval remain enterprise responsibilities.

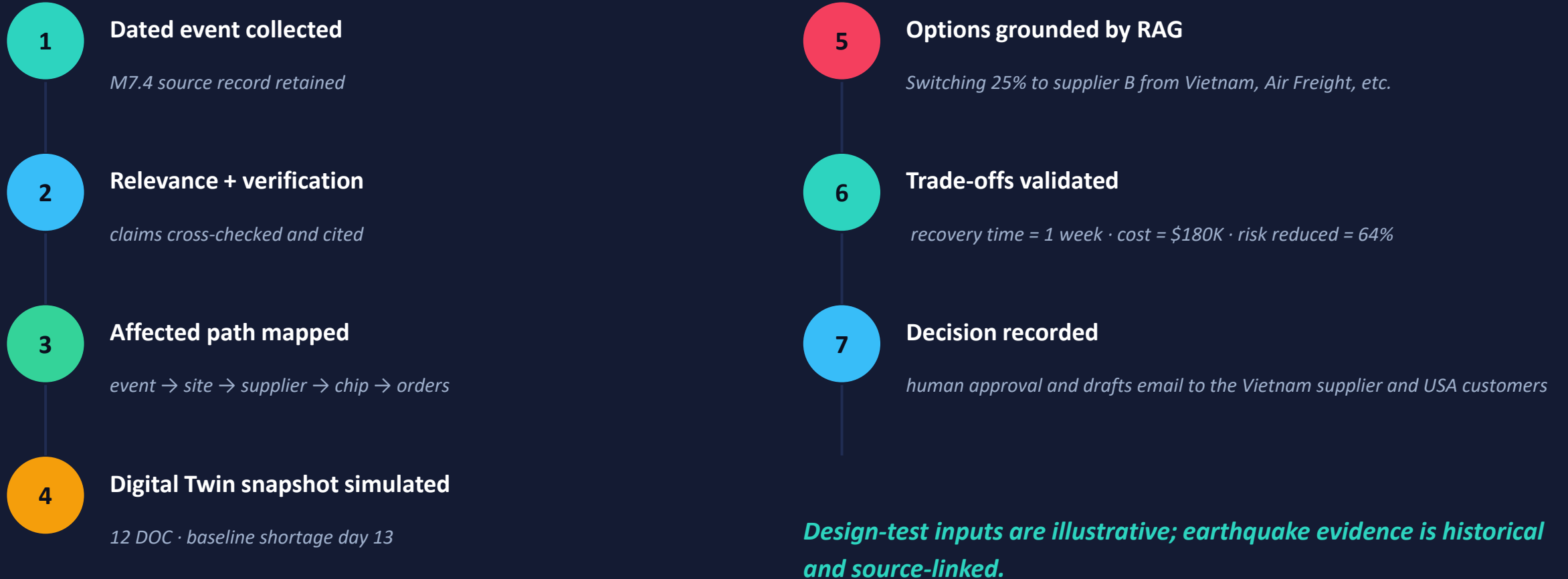
Evidence, exposure, impact, and action share one view



Top enterprise exposures			
EVENT	AFFECTED PATH	TRIAGE	IMPACT
Taiwan earthquake	Supplier A → chip → 7 orders	Critical	\$600K revenue at risk
Port congestion	Shanghai–LA route	High	6-day delay
Export restriction	Rare-earth supplier	High	+18% cost

Each alert separates the no-action baseline, projected mitigated outcome, and realized outcome—and links every figure to evidence and the twin snapshot.

Case Study: Taiwan earthquake's evidence to decision flow



Claims are designed to be tested

Pilot acceptance criteria

- Detection: $\geq 85\%$ of disruptions within 24 hours
- Verification: $< 2\%$ unsupported claims
- Mapping: $\geq 80\%$ correct affected supply chain paths connected to disruption
- Action: $\geq 90\%$ feasible recommendations
- Workflow: 50% faster alert-to-decision time





CHAINSILIENCE AI

Verified disruption evidence. Enterprise-specific impacts and tailored mitigations. Human-approved action.

Start counteracting supply chain disruptions today.

4 × Computer Science

Ka Woon Ho, Ke Xin Cao, Mo Xia, Andrew Li

1 × Rotman Commerce

Gary Leng

University of Toronto · Focus: SMEs, startups, and semiconductor supply networks · Functional prototype ready for demonstration.